PARISHRAM 2.0

CLASS 12TH 2026

HUMAN REPRODUCTION

ZOOLOGY | LECTURE-3

-PARAM SIR



TODAY'S TOPIC -

1. REVISION
2. SPERMATOGENESIS *
3. STRUCTURE OF SPERM *



REVISION



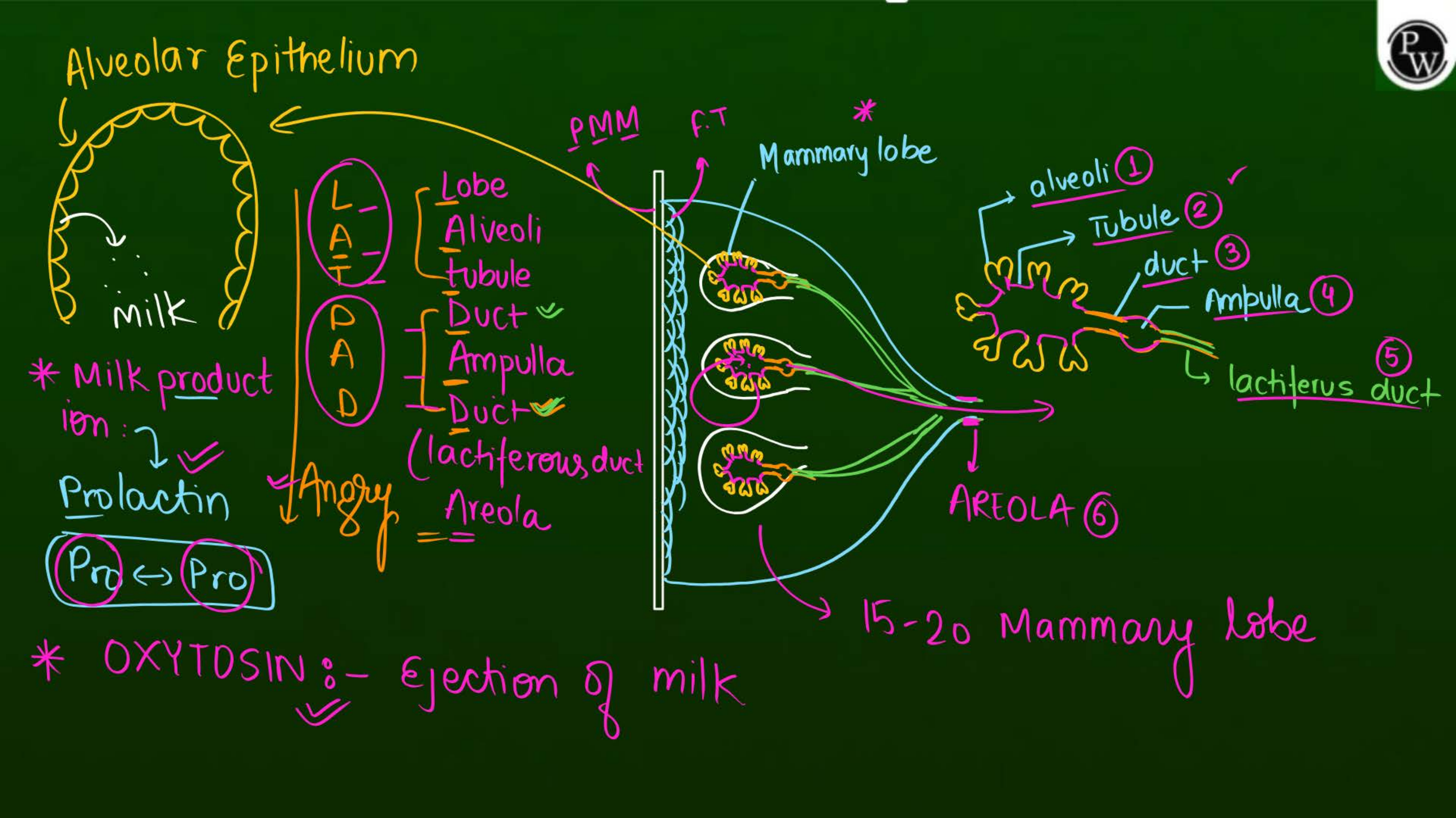
Mammary gland

- Function mammary gland is characterstic feature of ♀ mammals
- Paired st.
- located on Pectoralis major Muscles.

Mammary gland
- Undergo differentiation during pregnency
- Secrete milk after birth of baby.

G ✓
A ✓ *
M ✓
E ✓
D ✓

- First milk :- Colostrum → Antibodies → IgA * → Immunoglobulin


Alveolar Epithelium
Milk
* Milk production : ↓
Prolactin
Pro ↔ Pro
LAT
PAD
Lobe
Aliveoli
tubule
Duct
Ampulla
Duct
(lactiferous duct
Areola
Angry
PMM
F.T
*
Mammary lobe
alveoli (1)
Tubule (2)
duct (3)
Ampulla (4)
lactiferus duct (5)
AREOLA (6)
15-20 Mammary lobe
* OXYTOSIN :- Ejection of milk

NCERT-dig

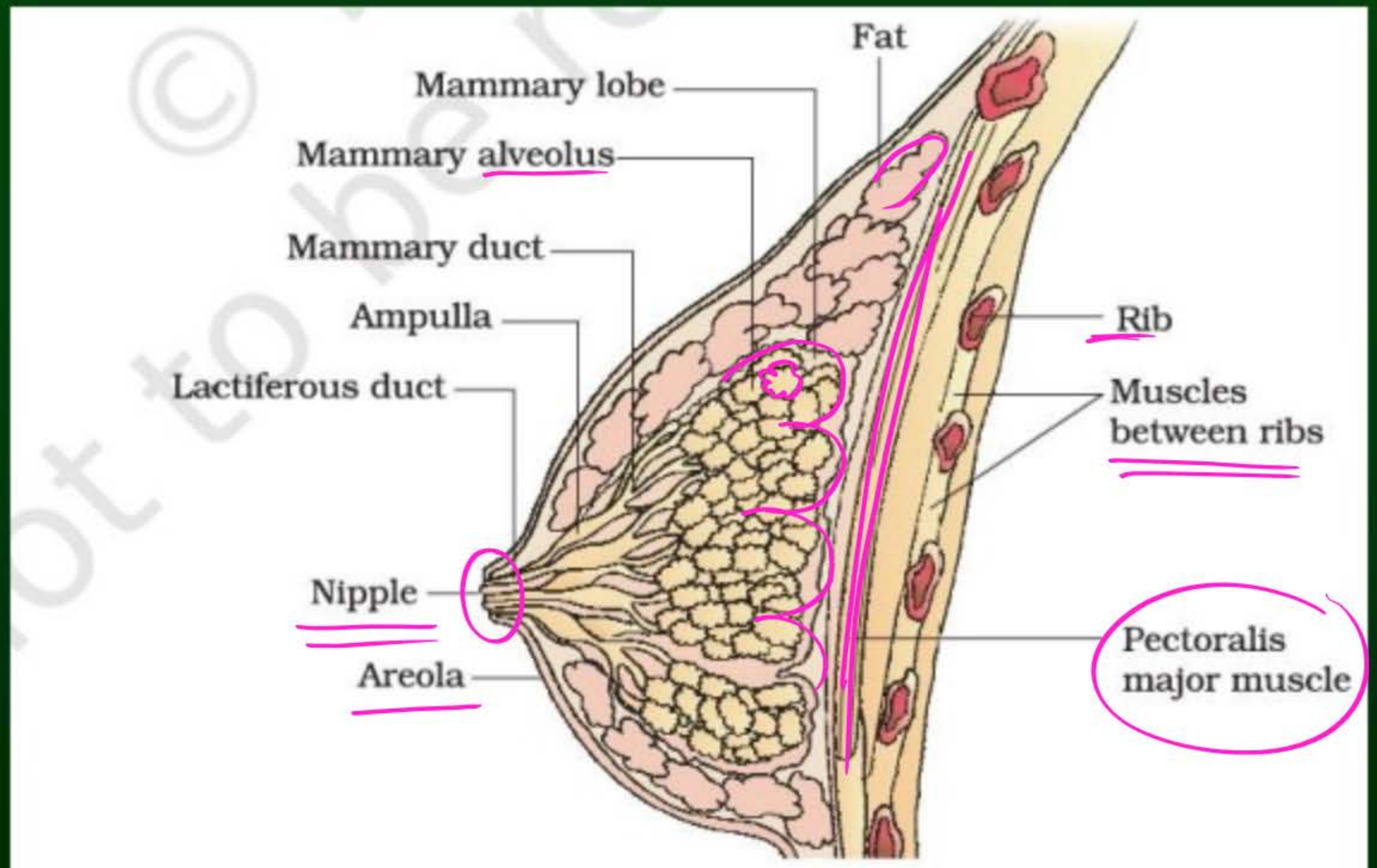


Figure 2.4 A diagrammatic sectional view of Mammary gland

L Mammary lobe
↓
A Mammary alveoli
↓
T Mammary Tubule
↓
D Mammary Duct
↓
A Mammary Ampulla
↓
D lactiferous Duct
↓
Angry AREOLA

Gametogenesis

↓

Spermatogenesis

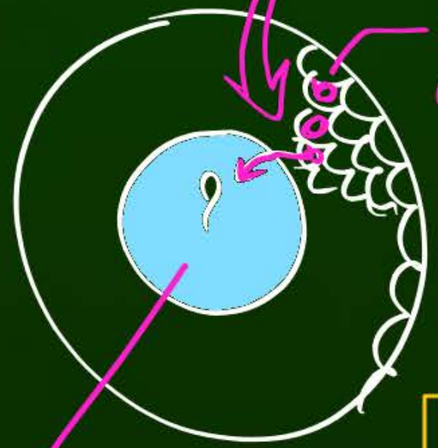
→ production of sperm

→ Start at puberty

→ Sperm production occur throughout life

SPERMIATION

Release of sperm in lumen of seminiferous tubule.



SPERMATO GENESIS

SPERMIO GENESIS

1. SPERMATOGONIA (2n) [44+XY]
   ↓ MITOSIS
2. 1° SPERMATOCYTE (2n) [44+XY]
   ↓ MEIOSIS-I (REDUCTION)
3. 22+X, 2° SPERMATOCYTE (n) 22+Y
   ↓ MEIOSIS II
4. SPERMATID (n)
   ↓
5. SPERM / SPERMATOZOA

22+X | 22+Y

NCERT-dig

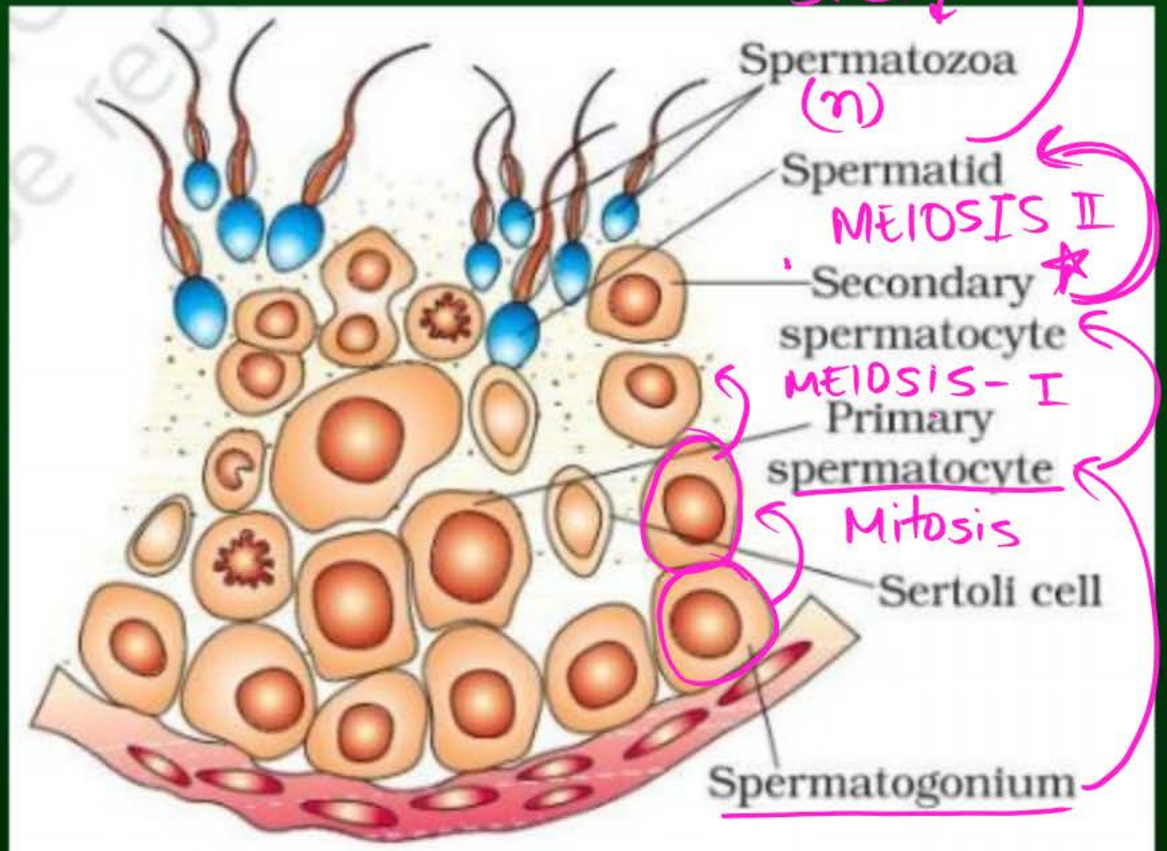


Figure 3.5 Diagrammatic sectional view of a seminiferous tubule (enlarged)

SPERMIOGENESIS

SPERMATO GENESIS

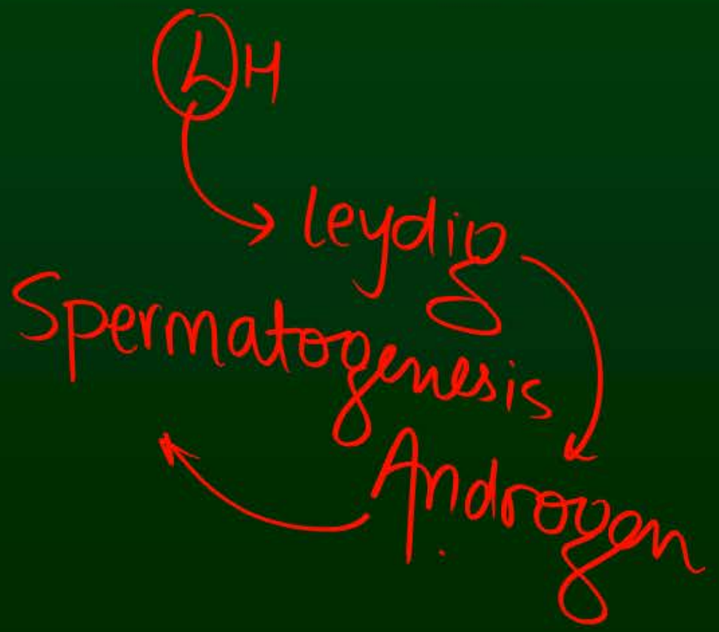
NCERT

EXTRACT

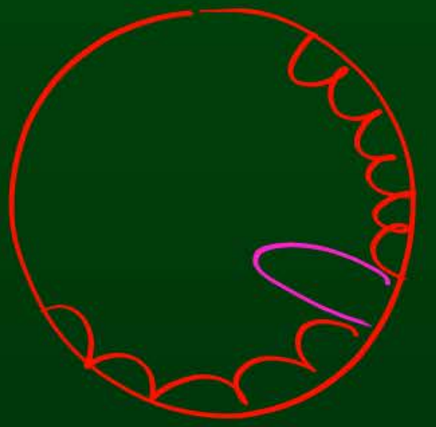
The primary sex organs – the testis in the males and the ovaries in the females – produce gametes, i.e, sperms and ovum, respectively, by the process called gametogenesis. In testis, the immature male germ cells (spermatogonia) produce sperms by **spermatogenesis** that begins at puberty. The **spermatogonia** (sing. spermatogonium) present on the inside wall of seminiferous tubules multiply by mitotic division and increase in numbers. Each spermatogonium is diploid and contains 46 chromosomes. Some of the spermatogonia called **primary spermatocytes** periodically undergo meiosis. A primary spermatocyte completes the first meiotic division (reduction division) leading to formation of two equal, haploid cells called **secondary spermatocytes**, which have only 23 chromosomes each. The secondary spermatocytes undergo the second meiotic division to produce four equal, haploid **spermatids** (Figure 3.5). *What would be the number of chromosome in the spermatids?* The spermatids are transformed into **spermatozoa** (**sperms**) by the process called **spermiogenesis**. After spermiogenesis, sperm heads become embedded in the **Sertoli cells**, and are finally released from the seminiferous tubules by the process called **spermiation**.

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Spermatogenesis starts at the age of puberty due to significant increase in the secretion of gonadotropin releasing hormone (GnRH). This, if you recall, is a hypothalamic hormone. The increased levels of GnRH then acts at the anterior pituitary gland and stimulates secretion of two gonadotropins – luteinising hormone (LH) and follicle stimulating hormone (FSH). LH acts at the Leydig cells and stimulates synthesis and secretion of androgens. Androgens, in turn, stimulate the process of spermatogenesis. FSH acts on the Sertoli cells and stimulates



EXTRACT



INTERSTITIAL SPACE
LAYDIG CELL

Hormonal Regulation

(2n) (n)
SPERMATOGONIA → SPERM

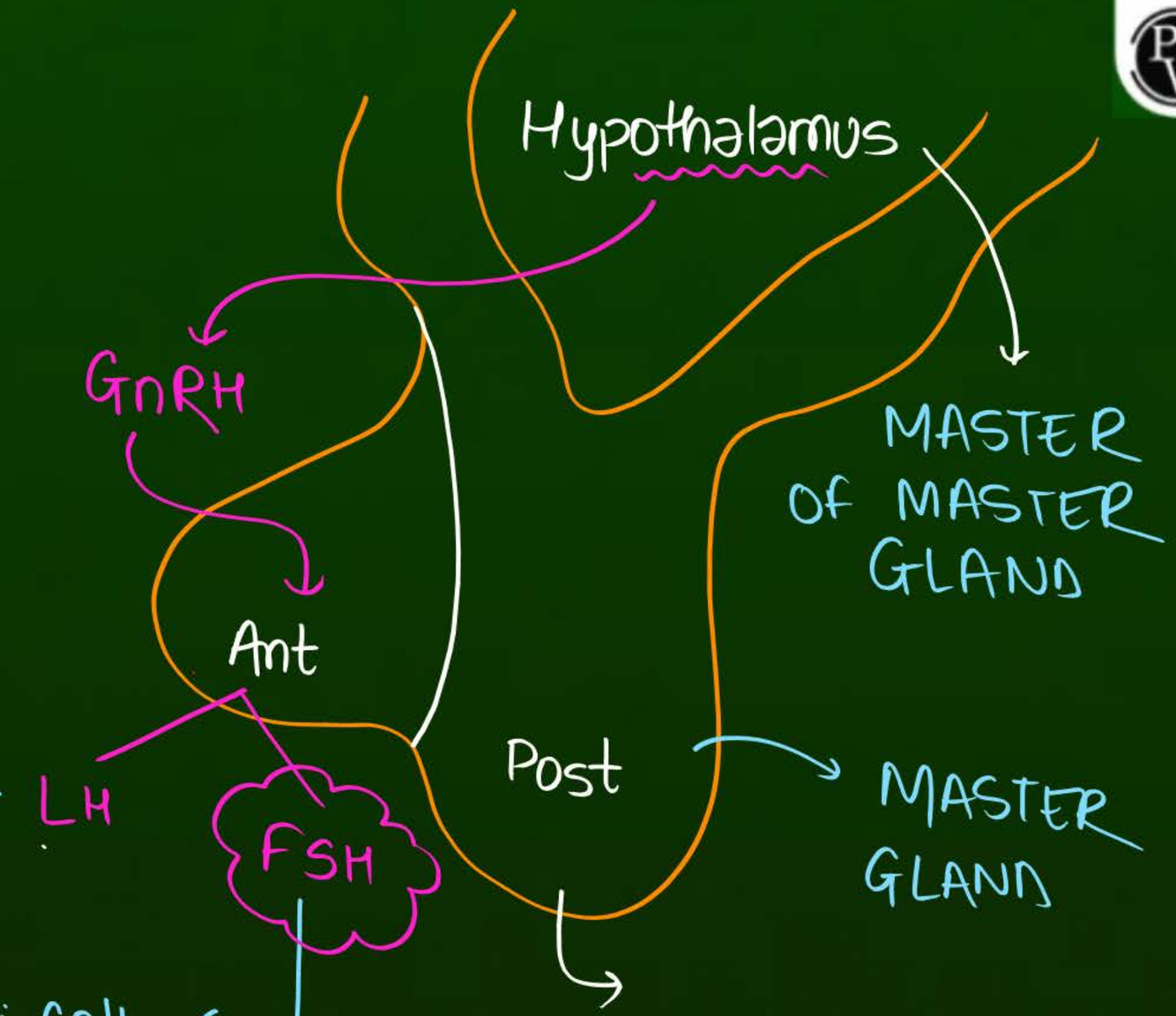
SPERMATOGENESIS

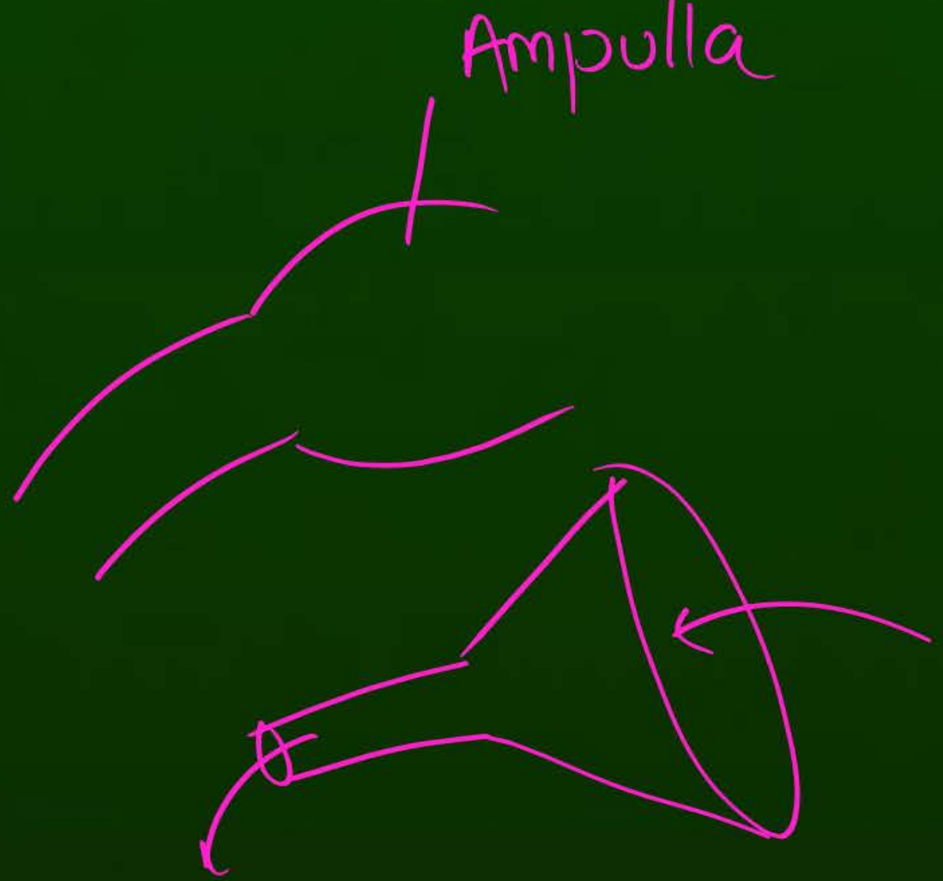
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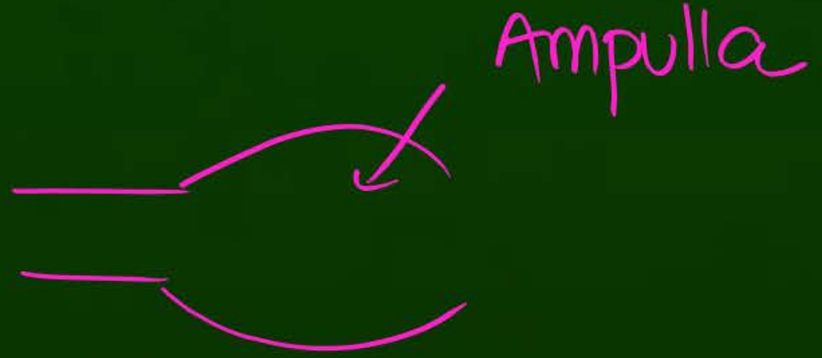
Androgen ← Leydig Cell ← LH

SPERMEIOGENESIS ← Sertoli Cell

SPERMADID → SPERM *
(n) (n)




Ampulla


Ampulla

NCERT

secretion of some factors which help in the process of spermiogenesis.

Let us examine the structure of a sperm. It is a microscopic structure composed of a **head**, **neck**, a **middle piece** and a **tail** (Figure 3.6). A plasma membrane envelops the whole body of sperm. The sperm head contains an elongated haploid nucleus, the anterior portion of which is covered by a cap-like structure, **acrosome**. The acrosome is filled with enzymes that help fertilisation of the ovum. The middle piece possesses numerous mitochondria, which produce energy for the movement of tail that facilitate sperm motility essential for fertilisation. The human male ejaculates about 200 to 300 million sperms during a coitus of which, for normal fertility, at least 60 per cent sperms must have normal shape and size and at least 40 per cent of them must show vigorous motility.

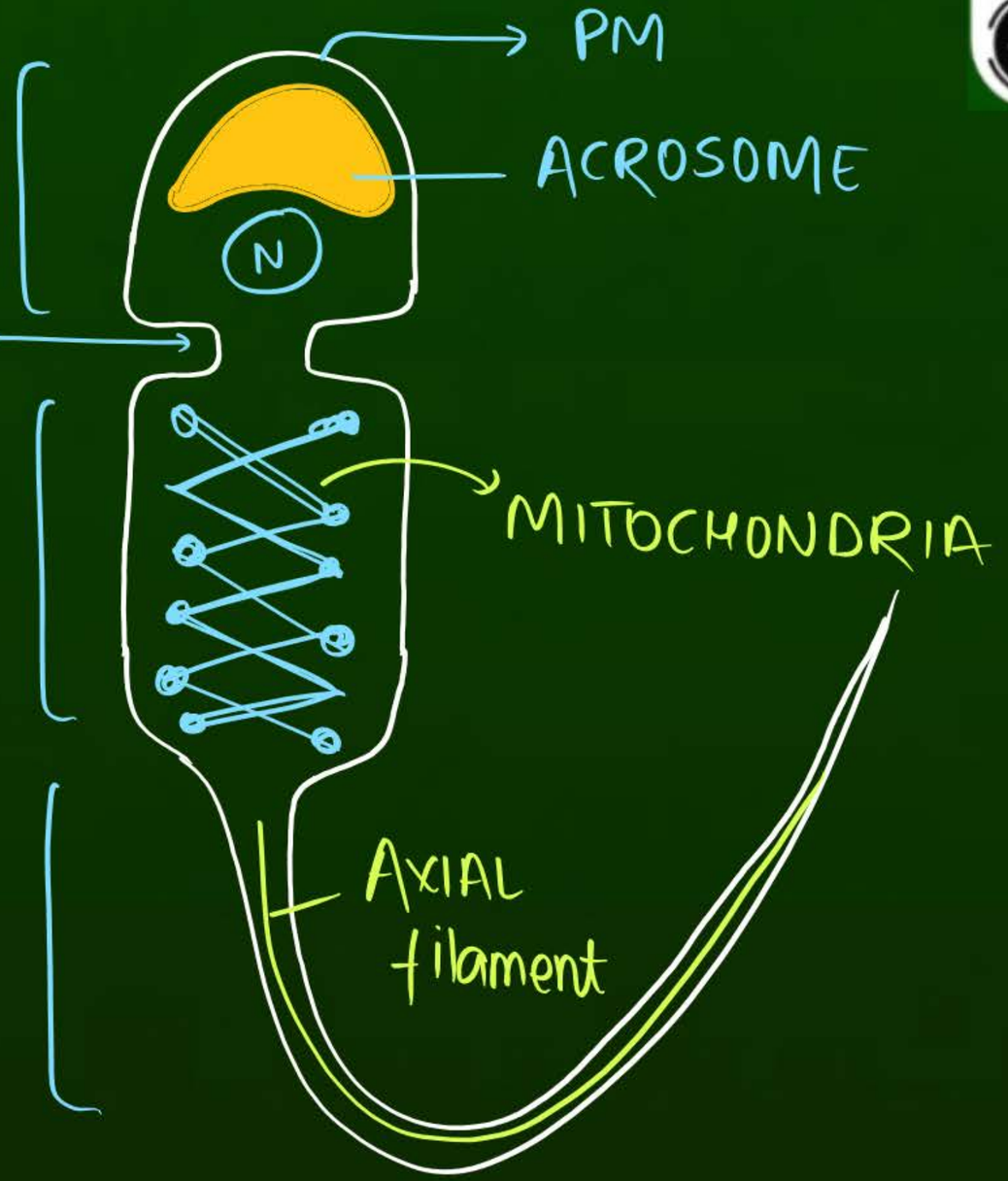
EXTRACT

SPERM

SPERM

- Body : Head, neck, middle piece & tail
- Acrosome - Caplike + ENZYME
- Nucleus - (n) and elongated
- MITOCHONDRIA - ENERGY
- HEALTHY :- 2.5 - 5ml : Semen

  [200 - 300] Million No.
- 60% of sperm should be perfect in shape and size → out of these 40% should VIGROUSLY ACTIVE | MOTILE



NCERT

EXTRACT

Sperms released from the seminiferous tubules, are transported by the accessory ducts. Secretions of epididymis, vas deferens, seminal vesicle and prostate are essential for maturation and motility of sperms. The seminal plasma along with the sperms constitute the **semen**. The functions of male sex accessory ducts and glands are maintained by the testicular hormones (androgens).

NCERT-dig

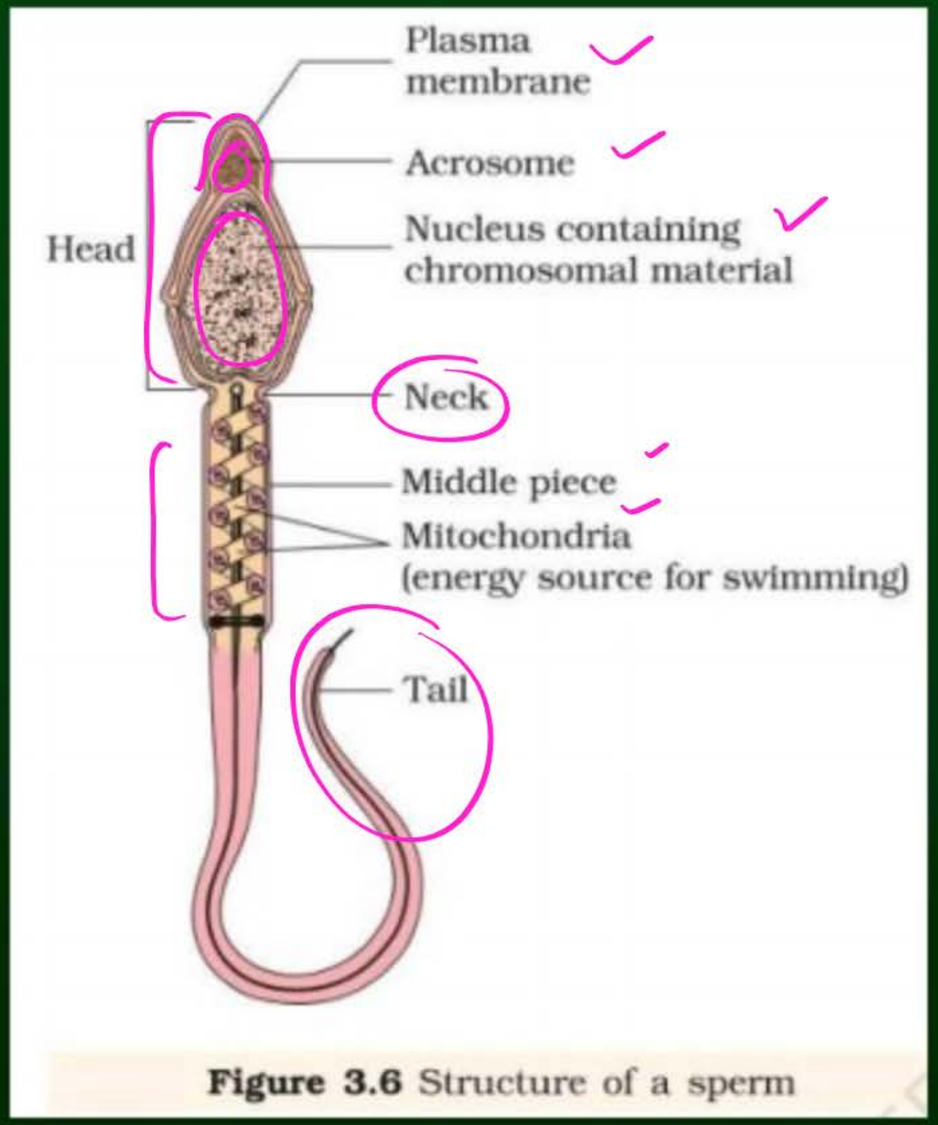


Figure 3.6 Structure of a sperm

THANK YOU